

Time: 90 Minutes

Maximum Marks: 100

General Instructions:

1. This paper is divided into four sections: A, B, C, and D.
 2. **Section A** consists of 25 Multiple Choice Questions (MCQs). Each question carries 2 marks. ($25 \times 2 = 50$ marks)
 3. **Section B** consists of 10 True/False and Fill in the Blanks questions. Each question carries 1 mark. ($10 \times 1 = 10$ marks)
 4. **Section C** consists of 5 Matching Type questions. Each question carries 2 marks. ($5 \times 2 = 10$ marks)
 5. **Section D** consists of 10 Short Answer and Case-Based questions. Each question carries 3 marks. ($10 \times 3 = 30$ marks)
 6. All questions are compulsory. Write neatly and number your answers correctly.
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Section A: Multiple Choice Questions ($25 \times 2 = 50$ Marks)

1. Which of the following correctly represents the hierarchy of data storage from smallest to largest?

- a) Bit → Byte → Nibble → Kilobyte → Megabyte
- b) Bit → Nibble → Byte → Kilobyte → Megabyte
- c) Byte → Bit → Nibble → Megabyte → Kilobyte
- d) Nibble → Bit → Byte → Megabyte → Kilobyte

2. Which component of the CPU performs arithmetic and logical operations?

- a) Control Unit
- b) Memory Unit
- c) ALU (Arithmetic Logic Unit)
- d) Register

3. The speed of a processor is measured in:

- a) Hertz (Hz)
- b) Bytes per second
- c) Pixels per inch
- d) Bits per second

4. Which of the following is an example of a utility software?

- a) MS Word
- b) Antivirus software
- c) Windows 10
- d) Adobe Photoshop

5. The binary equivalent of the decimal number 13 is:

- a) 1011
- b) 1101
- c) 1110
- d) 1001

6. The decimal equivalent of the binary number 1010 is:

- a) 8
- b) 9
- c) 10
- d) 11

7. Which of the following is a type of system software?

- a) Word Processor
- b) Spreadsheet
- c) Device Driver
- d) Presentation Software

8. The term 'GUI' stands for:

- a) General User Interface
- b) Graphical User Interface
- c) Graphical Universal Interface
- d) General Universal Interface

9. Which of the following storage devices has the FASTEST data access speed?

- a) Hard Disk Drive (HDD)
- b) Solid State Drive (SSD)
- c) DVD
- d) USB Flash Drive

10. The resolution of a monitor refers to:

- a) The physical size of the screen
- b) The number of pixels displayed on the screen
- c) The brightness of the screen
- d) The weight of the monitor

11. Which of the following is NOT a type of printer?

- a) Inkjet Printer
- b) Laser Printer
- c) Dot Matrix Printer
- d) LCD Printer

12. A device that connects multiple computers in a network and directs data packets between them is called a:

- a) Modem
- b) Router
- c) Repeater
- d) Hub

13. The protocol used for secure communication over the Internet is:

- a) HTTP
- b) FTP
- c) HTTPS
- d) SMTP

14. Which of the following is an example of cloud storage?

- a) Hard Disk
- b) Pen Drive
- c) Google Drive
- d) DVD

15. The process of converting data into a secret code to prevent unauthorized access is called:

- a) Decryption
- b) Encryption
- c) Compression
- d) Compilation

16. Which of the following is an example of an e-commerce website?

- a) Wikipedia
- b) Amazon
- c) Google
- d) YouTube

17. In MS Word, the feature used to create multiple copies of the same letter with different addresses is called:

- a) Copy-Paste
- b) Mail Merge
- c) Find and Replace
- d) Spell Check

18. In MS Excel, the function to find the largest value in a range of cells is:

- a) =MIN(A1:A10)
- b) =MAX(A1:A10)
- c) =SUM(A1:A10)
- d) =AVERAGE(A1:A10)

19. In MS PowerPoint, the shortcut key to start a slide show from the beginning is:

- a) F5
- b) F1
- c) F7
- d) Ctrl + S

20. Which of the following is a valid email address format?

- a) rahul@
- b) @[rahul.com](#)
- c) rahul@company
- d) rahul@company.com

21. A computer virus that disguises itself as a legitimate program is called a:

- a) Worm
- b) Trojan Horse
- c) Spyware
- d) Adware

22. The practice of sending fraudulent emails to trick people into revealing personal information is called:

- a) Spamming
- b) Phishing
- c) Hacking
- d) Cracking

23. Which of the following is a valid IP address?

- a) 256.256.256.256
- b) 192.168.1.1
- c) 300.100.50.25
- d) 192.168.1

24. The shortcut key to switch between open applications in Windows is:

- a) Ctrl + Tab
- b) Alt + Tab
- c) Shift + Tab
- d) Windows + Tab

25. Which of the following is NOT an example of a pointing device?

- a) Trackball
- b) Touchpad
- c) Joystick
- d) Barcode Reader

Section B: True/False and Fill in the Blanks (10 × 1 = 10 Marks)

A. True or False (Write 'True' or 'False')

26. A Nibble consists of 8 bits.

27. The Control Unit of the CPU directs and coordinates all the operations of the computer.

28. FTP is used for sending and receiving emails.

29. A plotter is an output device used to print large-format drawings and designs.

30. In a computer network, a server provides resources and services to client computers.

B. Fill in the Blanks (Write the complete sentence with the correct word)

31. The main circuit board of a computer that connects all components is called the _____.
32. The process of loading the operating system into the computer's memory is called _____.
33. The language consisting of only 0s and 1s that a computer understands directly is called _____.
34. The software program that manages files and folders on a computer is called a _____.
35. The shortcut key combination to force-quit an unresponsive program in Windows is _____.
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Section C: Matching Type Questions (5 × 2 = 10 Marks)

Match Column A with Column B. Write the correct letter next to the number.

Column A

Column B

- | | |
|----------|-----------------------------|
| 36. .exe | a. System file |
| 37. .sys | b. Image file |
| 38. .jpg | c. Audio file |
| 39. .mp3 | d. Executable file |
| 40. .pdf | e. Portable Document Format |

Answers format: 36-d, 37-a, etc.

Section D: Short Answer and Case-Based Questions (10 × 3 = 30 Marks)

41. The Binary Conversion Challenge (Case-Based)

A student is learning about the binary number system in computer class.

- a) Convert the decimal number 25 to binary. (1 mark)
- b) Convert the binary number 1110 to decimal. (1 mark)
- c) Why do computers use the binary number system instead of the decimal number system? (1 mark)

42. The File Management Puzzle (Application-Based)

Riya has created the following files on her computer:

- "my letter.txt"
- "science project.pptx"
- "class timetable.xlsx"
- "school logo.jpg"

- a) Which software will she need to open "science project.pptx"? (1 mark)
- b) Which file is an image file? (1 mark)
- c) Which file was created using a text editor like Notepad? (1 mark)

43. The Network Identification (Experimental Design)

A school has set up a computer network in its computer lab.

- a) What type of network (LAN, MAN, or WAN) is the school's computer lab network? (1 mark)
- b) Name one device that connects all computers in the lab to form a network. (1 mark)
- c) State one advantage of having a computer network in the lab. (1 mark)

44. The Software Classification (Conceptual Understanding)

A student identifies different types of software on a computer.

- a) Differentiate between system software and application software. (2 marks)
- b) Give one example of each. (1 mark)

45. The Cyber Security Scenario (Logical Reasoning)

Rahul receives a pop-up message on his computer saying "You have won a free iPhone! Click here to claim your prize."

- a) What type of cyber threat is this? (1 mark)
- b) What should Rahul do? (1 mark)
- c) Name one type of malicious software that could harm his computer. (1 mark)

46. The MS Word Formatting Question (Application)

In MS Word, a student wants to create a document with the following formatting:

- The title should be bold and centered
- The text should be double-spaced
- The page margins should be set to 1 inch on all sides

- a) Which toolbar option is used to make text bold and centered? (1 mark)
- b) Which feature is used to set line spacing to double? (1 mark)
- c) Where can the student find the option to change page margins? (1 mark)

47. The Excel Formula Challenge (Numerical)

In a Microsoft Excel spreadsheet, the following values are entered:

- Cell A1: 15
- Cell A2: 20
- Cell A3: 25
- Cell A4: 30

- a) Write the formula to find the average of these four cells. (1 mark)
- b) Write the formula to find the total of these four cells. (1 mark)
- c) What will be the result of the formula =A3-A1? (1 mark)

48. The Internet Concept (Conceptual Understanding)

- a) Differentiate between a web browser and a search engine. (2 marks)
- b) Give one example of each. (1 mark)

49. The Input-Output-Processing-Storage Model (Conceptual Understanding)

A student types a letter on the computer and prints it. Trace the flow of data through the computer system.

- a) Which device is used for input? (1 mark)
- b) Which component processes the data? (1 mark)
- c) Which devices are used for output and storage? (1 mark)

50. The Computer Hardware Identification (Application)

A student opens a computer cabinet and observes the following components: a large green board with many chips, a fan, a rectangular power supply, and a hard disk drive.

- a) Identify the large green board with many chips. (1 mark)
- b) What is the function of the fan inside the computer? (1 mark)
- c) What is the function of the power supply unit? (1 mark)